Ritvik Tak

AI Engineer | Computer Vision & Edge AI

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PROFESSIONAL SUMMARY

AI Engineer building real-time computer vision systems for retail analytics, from object detection and multi-object tracking to edge deployment on NVIDIA Jetson. Delivered footfall counting, dwell-time and zone-based analytics at 90–95% accuracy using Python, OpenCV and YOLO with ByteTrack and centroid tracking. Optimized models with ONNX and TensorRT and containerized with Docker for reliable inference on live RTSP camera streams.

TECHNICAL SKILLS

Computer Vision: YOLO, OpenCV, Object Detection, Multi-Object Tracking (ByteTrack, Centroid Tracking), Footfall Counting, Dwell-Time Analytics, Zone-Based Analytics, RTSP, FFmpeg
Edge & Deployment: NVIDIA Jetson (AGX, Nano), TensorRT, ONNX, Docker, JetPack, Real-Time Inference, Model Optimization
ML & Data: Python, PyTorch, NumPy, Pandas, Model Training, Dataset Annotation, Performance Validation
Backend & Databases: REST API Design, Flask, Spring Boot, MySQL, SQL
Languages: Python, Core Java, JavaScript
Tools: Git, GitHub, Linux, SSH, tmux, scp, Postman, remote.it, Makesense, VS Code
Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Low-Level and High-Level Design
Frontend (secondary): React.js, HTML5, CSS3

PROFESSIONAL EXPERIENCE

AI Engineer July 2026 – Present
Mialo Technology *Bangalore, India*

- Develop and test real-time retail analytics pipelines (RetailSense) in Python and OpenCV, delivering footfall counting, dwell-time and zone-based analytics across multi-zone deployments, with camera counts configured to each site's requirements.
- Deploy YOLO detection models on NVIDIA Jetson AGX and Nano edge devices, converting PyTorch weights to ONNX and TensorRT engines to run low-latency inference on live RTSP streams.
- Integrate ByteTrack and centroid tracking to keep object IDs consistent across frames, reducing duplicate counts and achieving 90–95% event and counting accuracy on live and recorded video.
- Troubleshoot RTSP connectivity and stream-path issues for Hikvision IP cameras using OpenCV and FFmpeg, keeping concurrent streams stable across sites with different camera counts.
- Containerize CV workloads with Docker and manage remote access to edge devices through remote.it for deployment, monitoring and debugging.
- Annotate datasets with Makesense, train custom detection models and validate accuracy on live and recorded video; expose results via REST APIs for downstream systems.

PROJECTS

Real-Time Footfall Counting System | Python, OpenCV, YOLO, NumPy, Pandas

- Built a people-counting system using YOLO on live and recorded video, counting unique individuals crossing a defined entry/exit line.
- Implemented centroid tracking with NumPy to prevent duplicate counts of the same person across consecutive frames.
- Logged timestamps, in/out counts and hourly totals with Pandas and exported summary reports to CSV.
- Tuned confidence thresholds and frame-skip intervals to balance real-time speed and accuracy on resource-constrained hardware.

User-Specific CRUD REST API | Java, Spring Boot, MySQL, Postman

- Developed a RESTful CRUD API with a layered Controller–Service–Repository architecture and a normalized MySQL schema; tested all endpoints with Postman.

EDUCATION

Bachelor of Science in Information Technology (B.Sc. IT) 2019 – 2023
MDS University, Ajmer, Rajasthan

TRAINING & CERTIFICATIONS

Scaler by InterviewBit – Software Engineering Program Jan 2023 – Present
1000+ hours across DSA, Low-Level and High-Level Design, Core Java, SQL and React.js, with Git workflows and peer code reviews.
Certifications: Core Java & OOP – Besant Technologies (2023); Database Management & SQL – Scaler (2023)
Languages: Hindi (Native), English (Professional Working Proficiency)